Model Fairness, Bias and Explainability Analysis

1. Objective

The objective of this task is to analyze machine learning model fairness, interpret predictions using SHAP, evaluate bias across sensitive groups and propose mitigation strategies.

2.Model Description

The model used in this project is **Logistic Regression**.

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts the probability of a class label using the logistic (sigmoid) function. The model estimates the relationship between input features and the target variable by fitting a linear decision boundary.

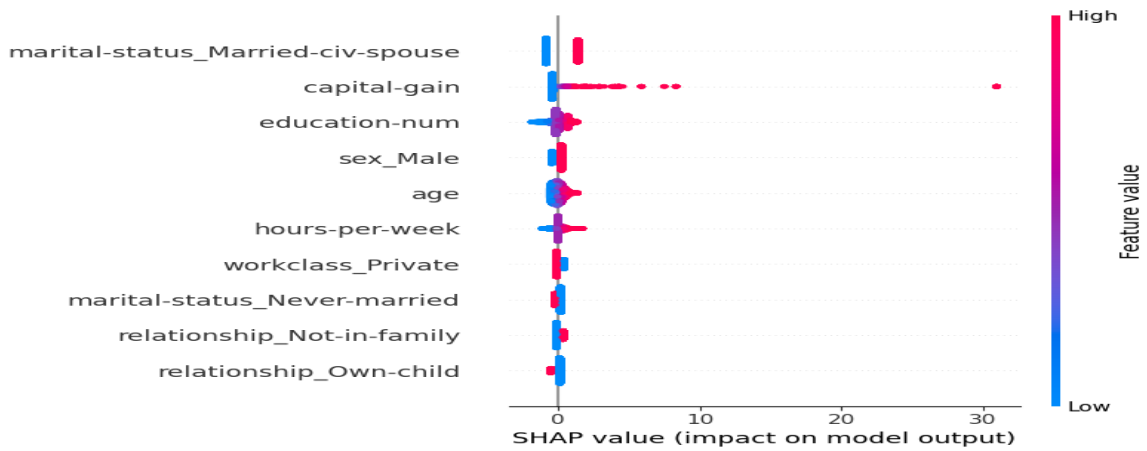
Logistic Regression is interpretable, computationally efficient, and suitable for understanding feature impact, making it appropriate for fairness and bias analysis.

3. Explainability Analysis (SHAP)

SHAP (SHapley Additive exPlanations) was used to interpret model predictions.

- Feature importance was computed.
- SHAP summary plot was generated.
- Individual feature contributions were analyzed.

The SHAP results shows which features most strongly influence the model's predictions.



4. Bias Analysis

Model performance across Gender groups:

Female Accuracy: 0.927
Male Accuracy: 0.814

The model shows better performance for Female samples compared to Male samples. This indicates performance disparity and potential bias across gender groups.

5. Mitigation Recommendations

To reduce bias and improve fairness, the following steps are recommended:

- Rebalancing the dataset
- Applying sample reweighting
- Using fairness-aware machine learning algorithms
- Adjusting decision thresholds
- Collecting more balanced data

6. Conclusion

The model explainability analysis using SHAP successfully identified important features influencing predictions. Bias analysis revealed performance disparity between gender groups. Appropriate mitigation strategies have been proposed to improve fairness and robustness.